1. Introduction

Amazon's Material Handling Systems (MHS) are undergoing a major transformation through the One-MHS initiative, a strategic effort to eliminate inefficiencies, improve system integration, and reduce reliance on vendor-dependent solutions. This initiative emerged as a response to challenges identified during Gen 11 facility launches, where disparate control systems, lack of standardization, and fragmented vendor solutions led to increased operational complexity, maintenance burden, and suboptimal product flow**.**

A critical pillar of this transformation is Project Catalyst, which shifts Amazon's Material Handling Equipment (MHE) architecture from vendor-controlled, isolated systems to an integrated, Amazon-owned framework. This new model establishes a standardized, hierarchical control system that facilitates real-time data exchange, predictive maintenance, and dynamic system optimization across fulfillment centers.

By implementing a vendor-agnostic approach, One-MHS retains control over critical operational parameters, ensuring consistency across all MHE systems while allowing flexibility in hardware selection. This document outlines the current state of MHS controls, the introduction of Project Catalyst, its necessity, architectural layers, and the roadmap for transitioning to a fully integrated One-MHS model.

2. Current State of MHS Controls Architecture

2.1 Current State with Vendor-Provided Solutions. Today, MHE systems operate in silos, limiting their ability to dynamically respond to changing fulfillment center conditions. Key challenges include:

a. Limited Inter-System Communication**:** Existing systems only transfer basic status information, restricting upstream/downstream adjustments.
b. Restricted Access to Data Tags & Values**:** Vendor systems do not surface all critical operational data to external sources, hindering Amazon's ability to define what "good" performance looks like.
c. Operational Inefficiencies**:** Without real-time performance insights, fulfillment centers struggle with product flow imbalances, increased stoppages, and ineffective jam recoveries.

3. Project Catalyst: One MHS Controls Architecture

3.1 What is Project Catalyst? Project Catalyst is an initiative developed by SPD-SEAD to create a standardized control layer that enables inter-system communication between MHE subsystems. It introduces a function block-based architecture that:

a. Monitors product flow conditions across fulfillment center operations.
b. Executes real-time adjustments by modifying speeds, gaps, and output rates.
c. Removes vendor constraints, allowing Amazon to incrementally take control of MHE system logic.
d. Ensures system continuity during communication failures, by shifting to a fallback logic layer.

This system is currently being tested at PDX8, with long-term ownership potentially transitioning to Amazon Warehouse Control System (AWCS)/Facility Management System (FMS), ensuring seamless governance under a centralized Amazon software system.

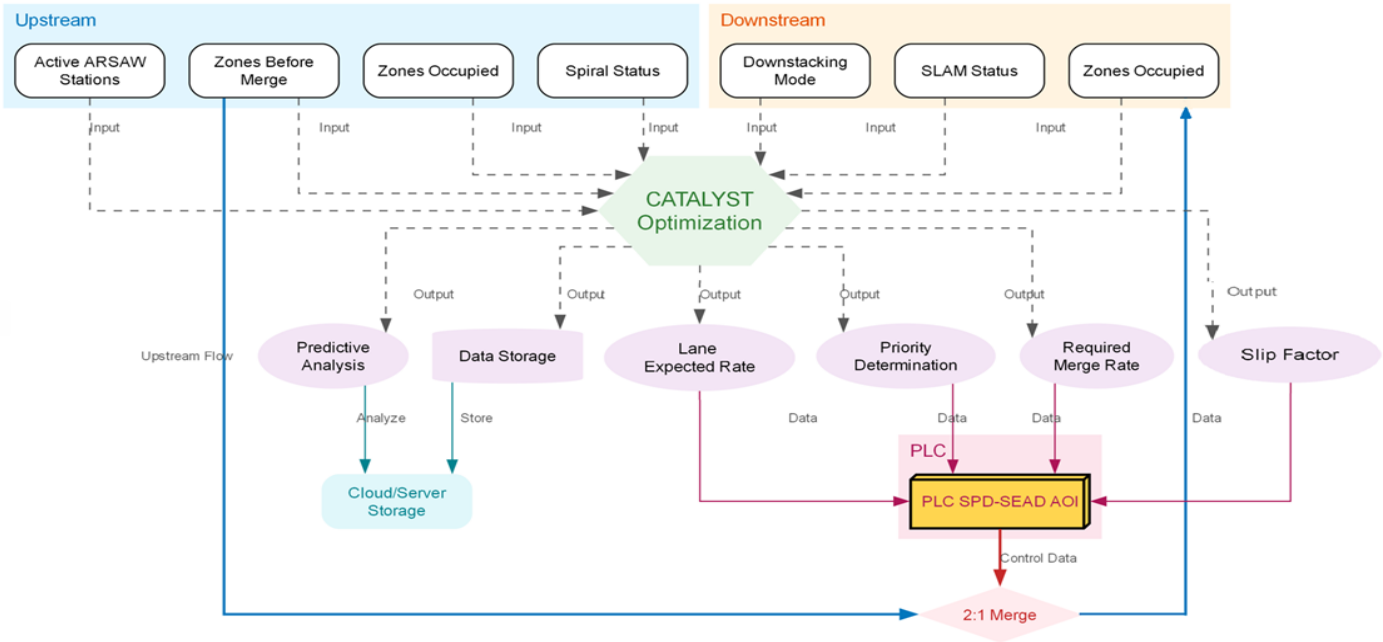


Figure 1: Project Catalyst

3.2 Why is Project Catalyst Needed?

During the Gen 11 launches, significant operational challenges arose, particularly within the AR collector belt system. These issues stemmed from non-standardized controls, site-specific modifications, and hardware variations, leading to performance inconsistencies. The System Performance & Development (SPD) team was frequently engaged in reactive support, developing stopgap solutions such as loop fullness thresholds and rate-limiting logic to maintain operational stability. However, maintaining these standards across multiple sites proved increasingly difficult as designs and hardware configurations varied.

To address these challenges, Automated Controls Qualification (ACQ) and Manual Controls Qualification (MCQ) checks were introduced to assess Programmable Logic Controller (PLC) code compliance with predefined standards. Higher compliance scores correlated with better-performing facilities, but as more Gen 11 sites launched with persistent issues, the demand for SPD and Software Engineering & Deployment (SEAD) support exceeded available resources. This led to the adoption of the KCVG collector loop method, where AWCS took control of the diverts instead of the site-level PLCs. SPD collaborated with Amazon Fulfillment Technologies (AFT) to develop a cloud-to-PLC communication solution, allowing cloud-based machine action requests to be executed by PLCs. The success of this approach culminated in Project Nemesis, which integrated controlled chaos principles with FMS logic to enhance AR collector loops by optimizing flow splitters and bi-directional systems.

Project Catalyst is the next evolution of this effort, designed to bridge control gaps, enhance real-time responsiveness, and establish a standardized MHE framework. Key objectives include:

a. Reducing Cloud-to-Hardware Latency: Current cloud-based execution delays introduce inefficiencies in merges and sorters, where even minor lags can disrupt product flow.
b. Maintaining Standardization While Enabling Adaptability: While standardization is essential, flexibility is needed to allow controlled experimentation and interplant data exchange.
c. Expanding Inter-System Communication: Vendor solutions provide only basic upstream/downstream messaging, which limits Amazon's ability to dynamically adjust flow conditions based on real-time data.

Example: Dynamic Rate Matching for Continuous Flow Optimization

Current fulfillment centers execute machine actions based on isolated inputs, rather than integrated upstream/downstream data flows. This results in inefficiencies such as gaps in product movement, suboptimal jam recovery, and blue-light conditions. Catalyst's interlinked control logic dynamically synchronizes machine centers, ensuring adaptive adjustments based on real-time processing conditions.

Example Scenario:

a. If a SLAM line is operating at reduced capacity due to a label roll change, the upstream zipper merge automatically reduces its rate to match.
b. This prevents excessive accumulation on conveyance lines, eliminating unnecessary stoppages.
c. Once the label roll is replaced, flow rates automatically adjust to restore full throughput, maintaining efficiency without manual intervention.
d. By integrating these dynamic control optimizations, Project Catalyst ensures Amazon's MHE systems operate as a unified, self-regulating network, capable of adapting to real-world conditions without reliance on vendor-specific control structures.

4. Architecture Layers

4.1 Hardware Layer

The Hardware Layer consists of essential MHE components, including photo eyes, motor-driven rollers (MDRs), variable frequency drives (VFDs), and other critical control elements. This layer is crucial for ensuring consistent, reliable performance across fulfillment centers.

To maintain system-wide standardization, One-MHS enforces strict hardware selection criteria, preventing uncontrolled site-level modifications that could impact flow performance. For example:

a. MDR cards without dip switches are required to prevent unauthorized field adjustments that could disrupt standard parameter settings.
b. All hardware selections must support Amazon-developed function blocks, enabling seamless integration with One-MHS controls.

By limiting variability in hardware components, Amazon ensures performance consistency, facilitates scalable deployments, and prevents non-compliant vendor solutions from affecting operations.

4.2 Interface Layer

The Interface Layer acts as a bridge between hardware and control logic, ensuring seamless communication between various system components. It translates diverse hardware communication protocols into a unified data stream, enabling:

a. Real-time gap adjustments based on upstream and downstream conditions.
b. Speed control commands to optimize product flow.
c. Reliable data integration with the Control Layer, facilitating system-wide coordination.

By standardizing interface protocols, One-MHS ensures consistent machine-to-machine communication, improving overall system responsiveness and reducing reliance on vendor-specific integrations.

4.3 Control Layer

The Control Layer consists of Programmable Logic Controllers (PLCs) and embedded control software, executing the real-time operational decisions that drive fulfillment center performance. Key functions include:

a. Requesting upstream/downstream adjustments to maintain steady flow.
b. Managing rate and gap controls to prevent congestion.
c. Executing system optimizations based on live performance data.
d. Maintaining fail-safe operations during cloud disconnection by running locally stored logic.

By enforcing PLC standardization and version control, One-MHS eliminates site-specific deviations, ensuring uniform performance across all fulfillment centers.

4.4 Supervisor Layer

The Supervisor Layer integrates AWCS, FMS, and Project Catalyst, enabling centralized real-time decision-making and system-wide coordination. Unlike traditional cloud-based orchestration, Project Catalyst operates within the building, allowing for:

a. Lower-latency execution of flow adjustments, reducing disruptions.
b. Full system integration, ensuring that all control decisions align with Amazon's automation framework.
c. Enhanced operational tools, including AI-driven predictive maintenance.

By embedding real-time control intelligence at the facility level, Project Catalyst minimizes reliance on cloud connectivity while improving MHE responsiveness and efficiency.

5. Standardizing Requirements for SSD vs ARS

One-MHS provides a unified control framework for both Sub-Same-Day (SSD) and Amazon Robotic Sortable (ARS) fulfillment models. While SSD operations require simpler configurations, ARS sites demand more advanced automation capabilities.

5.1 Sub-Same-Day (SSD) Requirements

Sub-Same-Day fulfillment centers require a streamlined approach with fewer control modules, focusing on efficiency and rapid order processing. Key requirements include:

a. 3-4 Core Modules: Minimal but essential components for effective operations.
b. X-to-1 Merge: Ensures product consolidation before sorting.
c. Accumulation Module: Manages buffer zones to avoid congestion.
d. Energy Management Module: Optimizes power consumption across material handling equipment.
e. Basic Control Features: Ensures standard functionality with minimal complexity.

5.2 Amazon Robotic Sortable (ARS) Requirements

Amazon Robotic Sortable sites require enhanced automation capabilities to manage complex order fulfillment operations. Key requirements include:

a. 7-8 Control Modules: Broader automation scope for advanced sorting.
b. Collector Loops: Ensures optimized package flow between sorter locations.
c. Bidirectional Systems: Allows movement in multiple directions to prevent bottlenecks.
d. Enhanced Integration Capabilities: Seamless connection to upstream/downstream MHE systems.
e. Data Collection & Scaling: Advanced tracking and real-time system monitoring.

5.3 Data Collection, System Optimization, & AI/ML Utilization

One-MHS relies on extensive data collection to enable real-time decision-making, AI-driven optimizations, and predictive analytics. To support this, robust data storage infrastructure and high-performance computing capabilities are necessary.

Key aspects include:

a. Project Catalyst will surface all module tags, ensuring integrated data streams for cross-team access and enabling software tools to better support operational needs.
b. Comprehensive Data Logging: Capturing real-time MHE performance metrics, system health data, and operator interventions.
c. AI/ML Model Integration: Using machine learning algorithms to optimize flow conditions, detect anomalies, and predict maintenance needs.
d. Real-time system monitoring, allowing for continuous performance tracking and anomaly detection.
e. Performance optimization capabilities, including the ability to push updates to base code remotely.
f. ChatSPD: AI-Powered Chatbot: A new, innovative chatbot designed to answer all MHE and One-MHS related queries, assisting operators, engineers, and decision-makers with instant data insights.
g. Scalable Data Storage Solutions: High-capacity cloud-based and on-prem storage to handle massive datasets, ensuring accessibility for analytics and historical benchmarking.
h. Decision Support Systems: Dashboards and interactive analytics tools that provide real-time system diagnostics and optimization recommendations.
i. Enhanced associate experience, reducing downtime and improving overall system reliability.

By integrating AI/ML-driven decision-making and real-time analytics, One-MHS ensures that decision-makers have actionable insights, improving efficiency, throughput, and system reliability across fulfillment centers.

5.4 Sub Same Day (SSD) vs. Amazon Robotics Sortable (ARS) Requirement Comparison

Fulfillment Type	Key Requirements	How Project Catalyst Supports
SSD	Basic MHE modules (X-to-1 merge, accumulation, energy management)	Provides pre-configured control modules for seamless MHE integration.
ARS	Complex collector loops, bidirectional systems, enhanced data collection	Implements AI-driven real-time flow adjustments, reducing manual intervention.

By leveraging standardized function blocks, One-MHS ensures that both fulfillment models benefit from predictable, scalable control solutions while eliminating vendor-driven inconsistencies.

6. Commissioning Efforts

6.1 Scaling Through 3P Vendor Support

As Amazon's One-MHS model successfully launches a building, stakeholders often push for rapid expansion, asking, *"Can we scale to ten buildings the following year?"* While this is ambitious, standardized control modules and pre-defined commissioning protocols enable efficient scaling.

To support this:

- Pre-developed hardware specifications and software modules reduce commissioning complexity.
- Qualified 3P vendors can implement software configurations and execute sorter commissioning without increasing Amazon's internal headcount.
- Regular feedback loops with 3P vendors allow Amazon to refine commissioning processes and improve performance.

By leveraging standardized templates and automation, Amazon can scale efficiently and predictably, minimizing resource constraints while maintaining operational consistency.

7. Appendix A: Automated Controls Qualification (ACQ) & Manual Controls Qualification (MCQ)

7.1 Ensuring Compliance in MHE Deployments

Vendors implement their standard control logic based on Amazon's MHE contractual performance requirements. However, site-level commissioning often introduces inconsistencies. ACQ/MCQ serve as a **compliance monitoring mechanism**, ensuring:

a. Vendor PLC code adheres to predefined standards without unauthorized modifications.
b. MHE systems operate at sustainable flow rates, rather than merely passing stress tests.
c. Remote and onsite verification streamline commissioning, reducing the need for manual interventions.

Future enhancements will expand ACQ scope to cover 100% of control tags, allowing:

a. Proactive monitoring of system health**.**
b. Real-time alerts for out-of-spec configurations**.**
c. Automated firmware and revision control updates**.**

8. Appendix B: Module Development

8.1 X-to-1 Merge: Future State – Catalyst IXD Fluid Load Lane

Problem**:** Traditional vendor-specific control layers introduce bottlenecks, requiring frequent site-level updates.

Solution**:** Catalyst eliminates reliance on vendor-driven configurations by:

a. Creating Amazon-controlled function blocks that standardize merge priority and release logic.
b. Removing unnecessary vendor modifications, ensuring alignment with upstream/downstream flow conditions.
c. Supporting remote updates, reducing downtime and engineering overhead.

8.2 1-to-X Sorter Optimization

Problem**:** Current sorter logic results in excessive product recirculation, overwhelming sorter lanes and increasing manual interventions.

Solution:

a. AWCS-integrated sorter intelligence dynamically adjusts accumulation zone capacity before releasing new product.
b. Reduces MaxRecirc occurrences, enhancing throughput and operational efficiency.
c. Prevents jackpot lane overflows, minimizing manual handling.

8.3 Collector Loop Logic Enhancements

Problem**:** Vendor-supplied collector loops operate at fixed thresholds, failing to adapt to real-time site conditions.

Solution:

a. Adaptive logic dynamically adjusts loop behavior based on package flow conditions.
b. Integrates AWCS and Catalyst control layers for consistent, system-wide optimization.
c. Enhances reliability by removing vendor-driven tuning inconsistencies.

8.4 Accumulation Conveyance: Product Flow Optimization

Problem**:** Static conveyance timers lead to inefficient product movement, increasing jams and reducing throughput.
Solution:
a. Real-time feedback loops between downstream and upstream conveyors enable dynamic release adjustments.
b. Prevents bunching issues, improving system stability.
c. Reduces start-stop inefficiencies, enhancing energy management.

8.5 Rate Matching Machine Centers: Pull the Rope Concept

Problem: Machine centers operate independently**,** failing to adjust rates based on real-time downstream conditions**.**
Solution:
a. Data-driven machine center adjustments synchronize product flow across the network.
b. Prevents accumulation overflows by adjusting merge rates dynamically.
c. Reduces downtime from jams, ensuring continuous throughput.

9. Appendix C: Frequently Asked Questions (FAQ)

9.1 What is Project Catalyst and why is it needed?

Project Catalyst is Amazon's **next-generation MHE control architecture**, designed to:
a. Eliminate vendor-driven inconsistencies by enforcing Amazon-controlled function blocks.
b. Enable real-time flow adjustments, optimizing throughput and reducing congestion.
c. Minimize site-level modifications, improving scalability and standardization.

9.2 How do 3P vendors fit into the One-MHS commissioning model?

3P vendors play a critical role in scaling deployment efforts, handling:
a. Software configuration and testing using Amazon's predefined standards.
b. Sorter commissioning and MHE validation, reducing Amazon's direct engineering burden.
c. Continuous feedback loops, improving commissioning methodologies over time.

9.3 How does ACQ/MCQ improve MHE standardization?

By enforcing real-time compliance checks, ACQ/MCQ ensures:
a. MHE systems adhere to predefined control logic**.**
b. Vendor modifications are detected and prevented**.**
c. Long-term system health monitoring and predictive maintenance capabilities**.**

9.4 How does AI/ML enhance One-MHS and Project Catalyst?

AI/ML is a key component of One-MHS, allowing for:
a. Automated real-time decision-making based on MHE system data.
b. Predictive maintenance models, reducing downtime and optimizing equipment performance.
c. ChatSPD: AI-powered chatbot, providing real-time insights on MHE operations and answering system-related questions.
d. Data-driven system optimizations, ensuring continuous learning and enhancement.

9.5 What are the data requirements for AI/ML-driven decision-making?

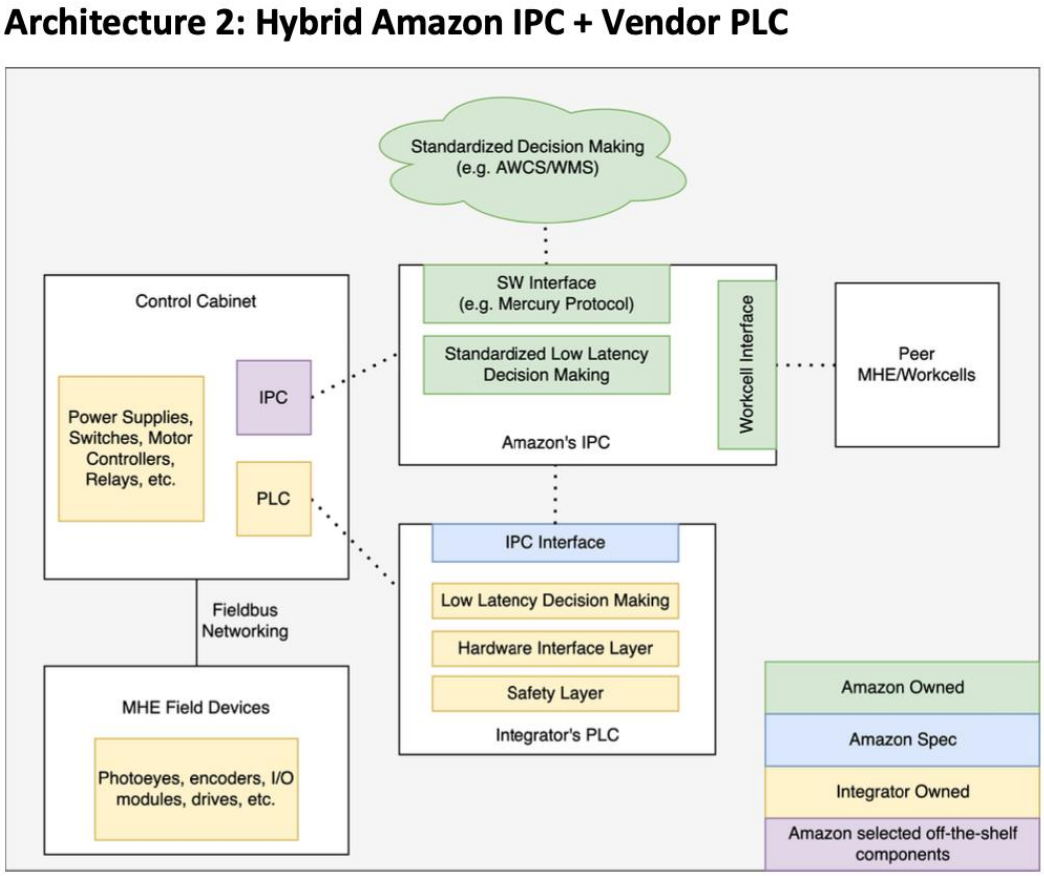
One-MHS relies on extensive data collection and storage to enable intelligent decision-making. The system requires:
a. Comprehensive data logging across all MHE components.
b. High-capacity cloud storage to support historical data analysis and machine learning training.
c. Integration with Project Catalyst, surfacing all module tags and enabling cross-team data accessibility.
d. Real-time system monitoring, capturing operational anomalies and improving process efficiencies.

9.6 What is ChatSPD and how does it support MHE operations?

ChatSPD is an AI-powered chatbot designed to:
a. Provide instant answers to MHE and One-MHS-related questions.
b. Assist engineers, operators, and decision-makers with real-time data insights.
c. Leverage AI-driven analytics to recommend optimizations and troubleshooting solutions.

9.7 How does One-MHS enhance associate experience?

By implementing real-time monitoring, AI-driven optimizations, and proactive maintenance, One-MHS:
a. Reduces downtime, ensuring smooth fulfillment operations.
b. Enhances system reliability, minimizing unexpected failures.
c. Improves system efficiency, allowing associates to focus on high-value tasks instead of manual troubleshooting.

9.8 How can Module develop be expediated?
a. Add additional Engineers to development efforts.
b. Increase collaboration efforts with other Amazon teams.
c. Utilizing software tools like Copia that handles revision controls and joint code development

10. Appendix D: Architecture 2